

Align site specific Induction: Tunnel



Accessing the Tunnel



Tunnel Training Requirements

All Operatives working in the Tunnel (Past Ring 100 up to & including the TBM) must have the following:

1. A Valid **CSCS Card**;
2. Completed the **Align Project & South Portal Inductions**;
3. A Valid **Safety Critical Medical (SCM)** Certificate;
4. A Valid **Confined Space Entry** Card (**6151-02**);
5. A Valid **Tunnel Entry** Certificate (**TSTS**), and
6. Completed this **Induction**



Item 1 to 5: **must** be uploaded on Biosite by your management to allow you the access through the tunnel's turnstiles

Item 6: will be awarded by the Traffic Controller in duty at the end of this Induction

Mandatory TComet Tracker

Once your Induction complete, you will be provided with a **Tcomet tracker**.

Please ensure that you have securely attached the **Tcomet Tracker** to your **helmet** or attached it with your **Badge**

These tracker will:

- Give you access to the Tunnel area through the turnstile
- Track your presence (or not) in the **tunnel in the event of an Emergency**

IT COULD SAVE YOUR LIFE



Mandatory TComet Tracker

- It is a responsibility of the owner to keep the tracker and maintained in good condition.

In case of lost or broken tracker:

- 1 Tracker lost – 1 Day of bonus removed
- 2 Tracker lost – 2 Days of bonus removed
- 3 Tracker lost – 3 Days + Verbal warning



Tunnel Workers Access Procedure

ALL Operatives working in the tunnel **MUST**:

1. Attend and sign onto a **SHIFT BRIEFING**;
2. Read and sign on to all relevant **RAMS** and **PROCEDURES** for their work;
3. **'TALLY IN' :**
 - Pass through the Electronic Turnstile adjacent to the Logistics Room to enter the Tunnels area
 - Activate the reader by scanning your **tracker**
 - Open the turnstile by scanning your **HS2** or **C1.Align Badge**
 - Access to the Tunnel & TBM is via a dedicated **PERSONNEL SERVICE VEHICLE**.
4. **'TALLY OUT':**
 - Pass out of Electronic Turnstile adjacent to the Logistics Room by scanning
 - Activate the reader by scanning your **tracker**
 - Open the turnstile by scanning your **HS2** or **C1.Align Badge**

Tunnel Special Access - 1

Walking to your workplace

If you are walking to your workplace from the turnstile, you need to report at the logistic cabin and ask a self-rescuer. This SR will need to be brought back to the cabin when going out or handed over to a PSV driver.



“Jumping” at the back of an MSV

If you **urgently** need to access one tunnel out of the PSV schedule, you can report at the logistic cabin and ask to go in with the next MSV. The traffic controller will confirm the next availability and inform the driver you will join him.

Before entering the MSV rear cabin, confirm everything is ok with the driver and inform him of where to stop for you

Tunnel Special Access - 2

Working on the invert

If you work in the Tunnel Invert, you must:

- a. Obtain an **INVERT WORK PERMIT** from the Traffic Controller
- b. Receive and keep safe possession of a **SELF-RESCUER** for the duration of their Works
- c. Obtain **Amber Flashing Beacons** and place them 50m In-Bye & Out-bye of your working zone
- d. Close the **INVERT WORK PERMIT** at the logistic cabin once works are complete

Note: An **INVERT WORK PERMIT** is automatically closed at the end of the shift. It must be requested by the next shift team if works are carrying on 24/7



Personnel Service Vehicle (PSV) Multi-Service Vehicle (MSV)

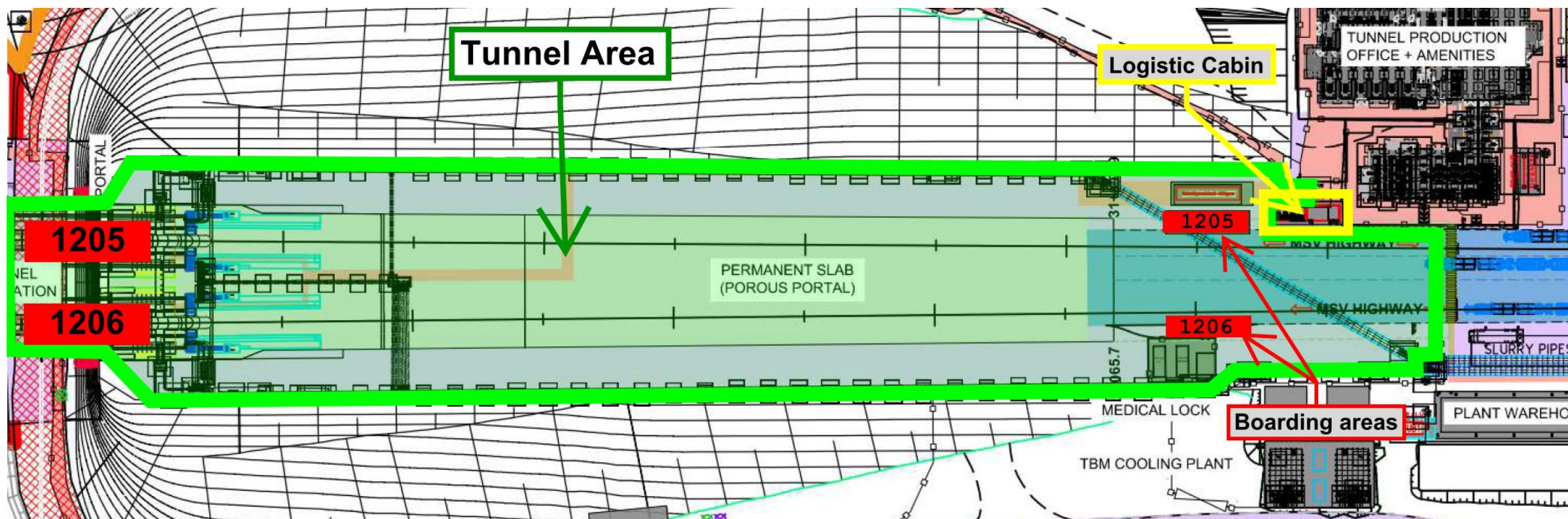
When in a vehicle:

- a. Wear a **SEATBELT** at all times
- b. Do not **REMOVE SEATBELT** or **OPEN THE DOOR** until the Driver has signalled the Vehicle is stopped (1 no. Long horn blast)

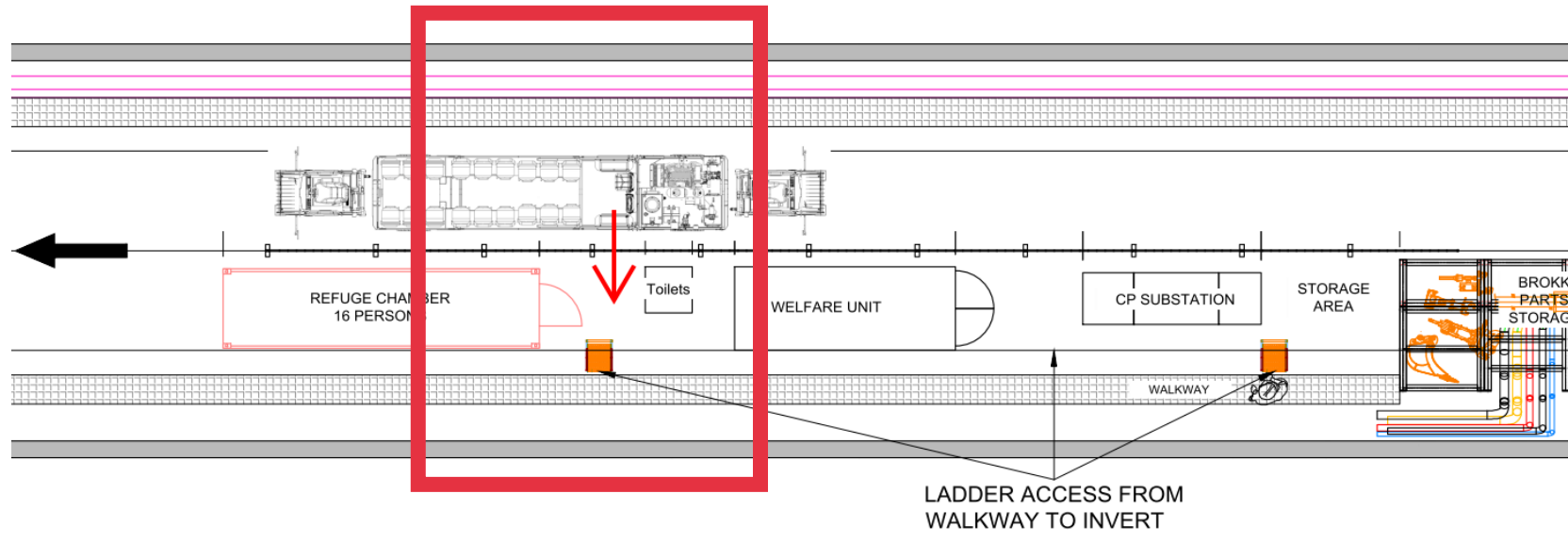
Note:

- It is **not allowed** to eat or smoke in the vehicle
- Rubbish **must** be placed in the skip if in a PSV or taken away if in an MSV
- It is not allowed to remove a Self-Rescuer from the vehicle, except in case of Emergency

Personnel Carrier – Embarkation on Surface



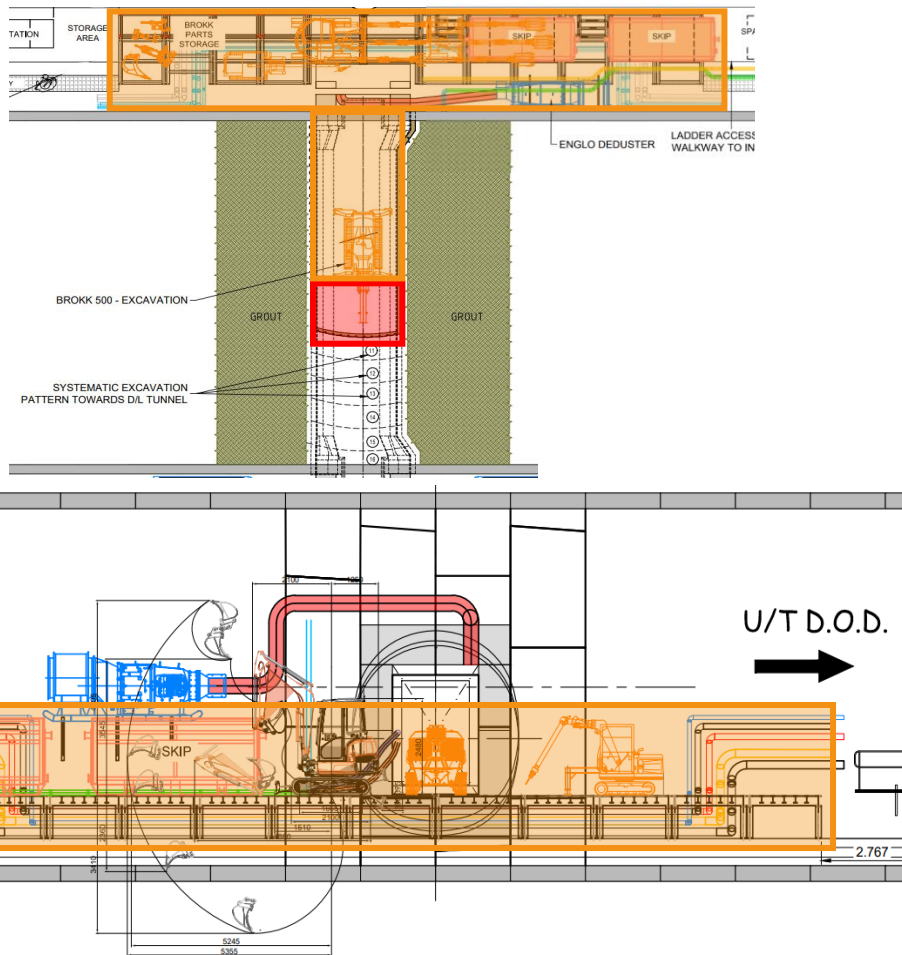
PSV – Embarkation at CPs



Access to CPs

- Due to the nature of the CP works and the space available access to the working area is strictly controlled
- The Working Platform is classed as a Restricted Access area
- To gain access to the working platform permission must be gained from the supervisor/Lead Miner
- There is to be no access into the cross passage during mining operations without the express permission of the Lead Miner
- The area in front of the mining operations is classed as an Exclusion Zone. There is no access to this area under any circumstances
- The following slides will explain this in greater detail

Restricted & Exclusion Zones



Plant operating:

- BROKK
- Excavator
- Sprayed concrete robot
- Concrete pump

Set-up:

- In the **Tunnel**
- On the platform
- In the **Tunnel**
- On the invert

-  Exclusion zone
-  Restricted zone

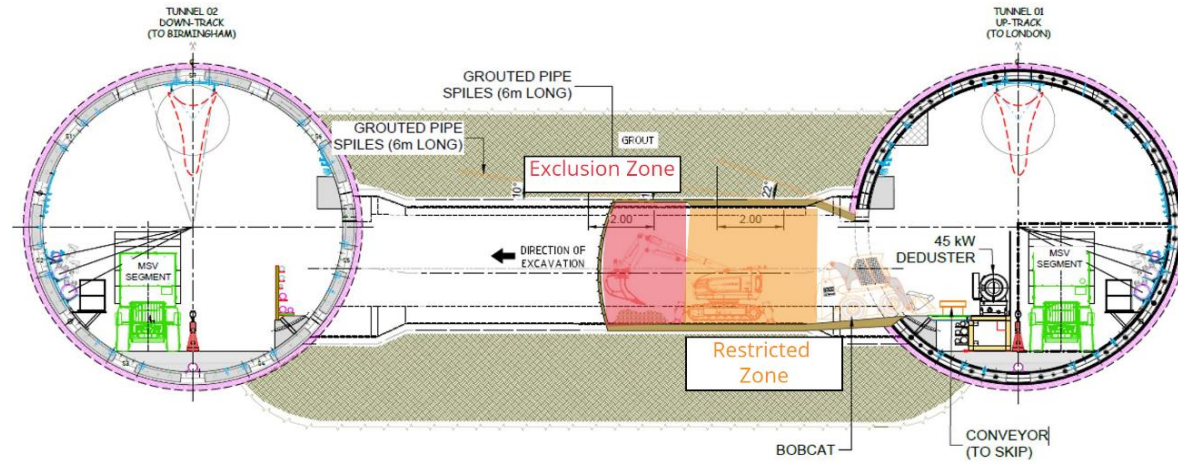
Key workers:

- BROKK driver
- Excavator driver
- Nozzleman
- Engineer



+ **Lead miner/Nozzleman** to allow needed workers for each activity

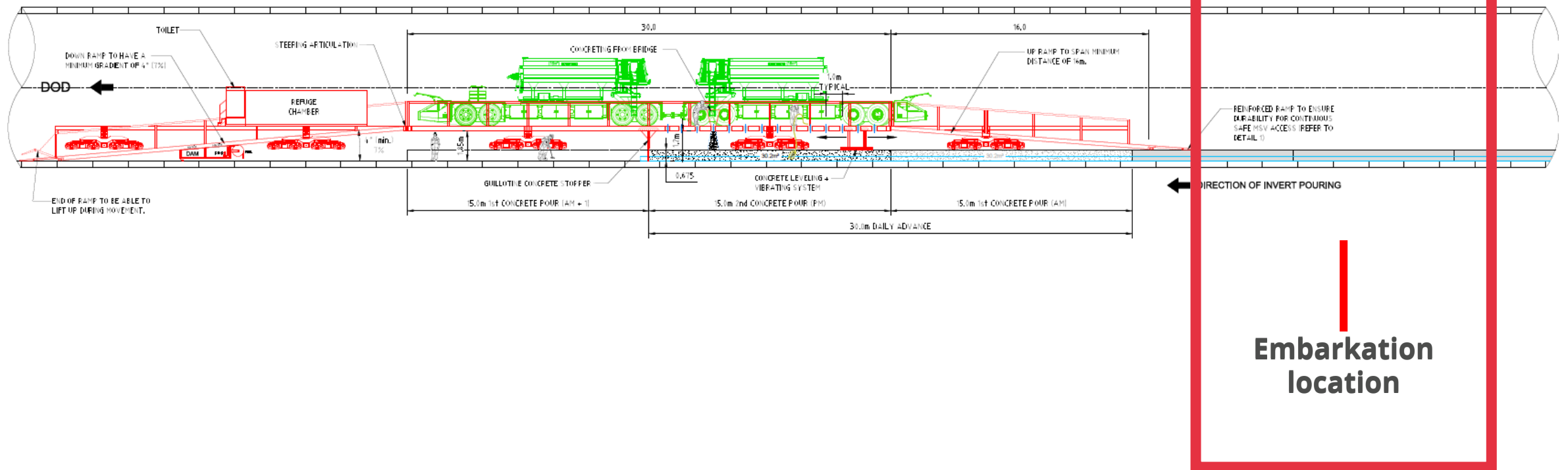
Exclusions and Restricted Zones



- Authorised workers
- Restricted access with strict access procedures in place during **spiling, excavation** and **spraying operations**
- Physical barriers in place (bungee cord or similar)

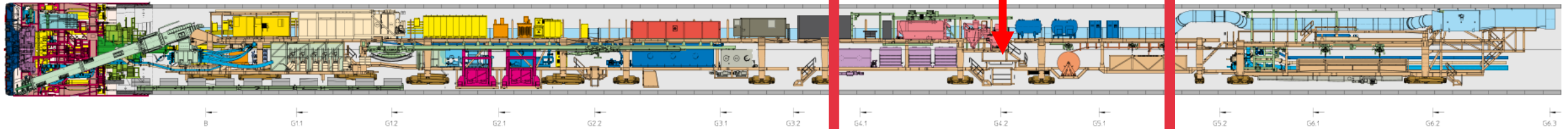
NO ACCESS
Under open ground
OR
Under fresh SCL before reaching **1 MPa**

PSV- Embarkation at the Bridge



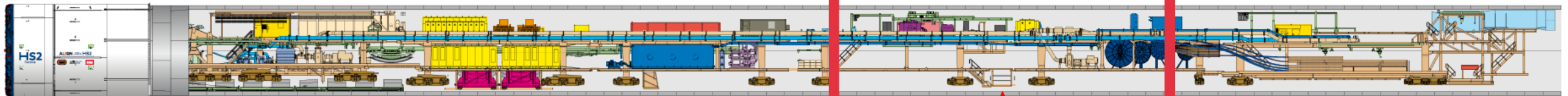
PSV- Embarkation on TBM

RIGHT SIDE



Embarkation location

LEFT SIDE



Driving in the Tunnel

- **ALWAYS** ensure you have received authorisation and training prior to driving any vehicle in the tunnel.
- **NEVER** operate a buggy without holding a buggy licence
- **ALWAYS** maintain 20 metres from the vehicle ahead (10 rings)
- **ALWAYS** wear a seat belt, and ensure your passengers are wearing theirs.
- **NEVER** use a phone or device while operating any vehicle, machine or tool.
- Speed Limits are **20mph** and **8mph** when passing working Cross Passages or Obstacles / Hazardous Areas

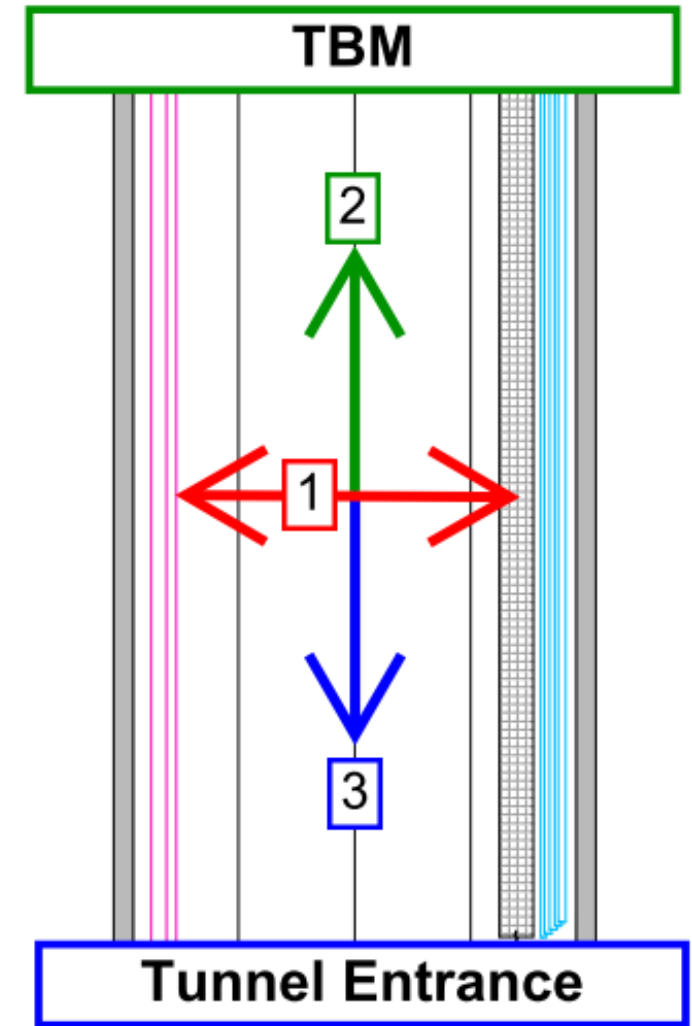


Any breach to the outlined rules will result in the operative being dismissed from site.

MSV Horn Blast

Horn Blast Signals:

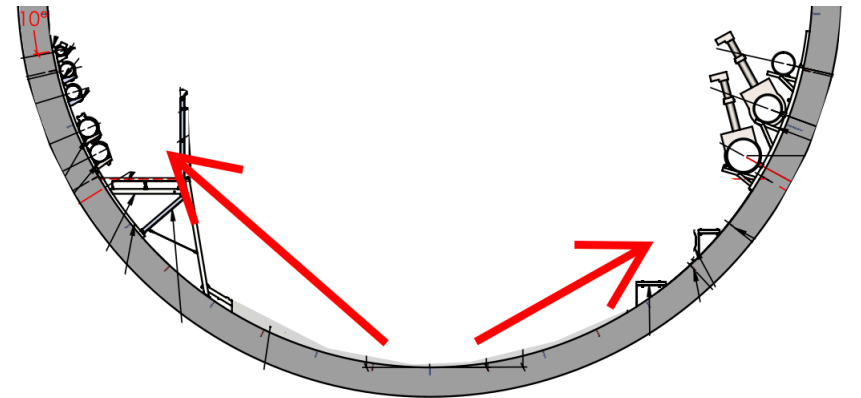
- **1 Long Horn Blast** – The MSV has come to a complete **STOP**
- **2 Horn Blasts** – The MSV is about to move in the direction of the TBM (**IN-BYE**)
- **3 Horn Blasts** – The MSV is about to move in the direction of the Pre-cast Factory (**OUT-BYE**)



Tunnel MSV Operations – Invert Slab

Working in the Tunnel Invert (Subject to Permit)

- a. When an MSV approaches work area (demarcated by Amber Beacon) – **Horn Blast & Slow Speed**
- b. All Operatives are to move to nearest ladder to the **Tunnel Walkway** or **Survey bench** & leave the **Tunnel Invert**
- c. MSV will stay at **Slow Speed** until all Operatives are clear and then may return to **Fast Speed**
- d. MSV at end of work area (other Amber Beacon) – **Horn Blast** & Operatives may return to Tunnel Invert



Tunnel MSV Operations – Invert Bridge

1. MSV Operation Over Invert Bridge:

- a. Drive slowly over Invert Bridge and obey all horn signals (**MAX 8 MPH**)
- b. Only **1 PRODUCTION MSV** can move **IN-BYE** of the ISCB at a time
- c. The Personnel Carrier / Tunnel Bus will move **IN-BYE** of the ISCB if directed

2. MSV Operation in TBM:

- a. Drive slowly into and out of TBM and obey all horn signals (**MAX 8 MPH**)
- b. Operator is not move MSV if personnel in TBM are in Invert

3. MSV Operation in Tunnel (Past CP, Pump Location or where Operatives are working:

- a. Drive slowly past working area and obey all horn signals (**MAX 8 MPH**)

Tunnel MSV Operations – In the TBM

Working in the TBM Invert

- a. When an MSV approaches Back-up Gantry 6 – **Horn Blast & Slow Speed**
- b. If Operatives need to continue undertake work in **TBM Invert**, MSV will **STOP** at Back-up Gantry 6
MSV is not allowed to move if anyone is standing in the invert
- c. MSV will **ONLY** enter Back-up Gantry 6 once Operatives have left the **TBM Invert** – directed by the **Lead Miner**
- d. When MSV is in TBM, **TBM Invert** is an **Exclusion Zone** unless Authorised by **Pit Boss**

Tunnel MSV Operations – In the TBM

Working in the TBM Invert

- a. When an MSV approaches Back-up Gantry 6 – **Horn Blast & Slow Speed**
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Main Risks and Hazards



Moving Plant



Manual Handling



Lifting



Contamination

Don't operate if you are not **TRAINED TO DO SO**

Ask your **SUPERVISOR** if you are unsure

If it **FEELS UNSAFE IT PROBABLY IS**

Spillage

- **In case of Spillage:**
 - Assess the risk and inform your **SUPERVISOR**
 - Wear the correct **PPE**
 - **CONTAIN** the Spillage
 - **ISOLATE** the Source
 - **CLEAN** up





Emergency Procedure

How to Raise the Alarm & Emergency Stop

Emergency Stoppage

1. **LOCATE** the **EMERGENCY STOP BUTTON**
2. **PUSH AND TWIST** the **EMERGENCY STOP BUTTON** in case of any emergency, to stop all moving parts and motors
3. **RAISE** the **ALARM**



How to Raise the Alarm

1. **LOCATE** the **FIRE and EVACUATION ALARM POINTS**
2. **PUSH** the **ALARM BUTTON** alerting the tunnel SIC and personnel that there is an incident
3. **EVACUATE** the area if required



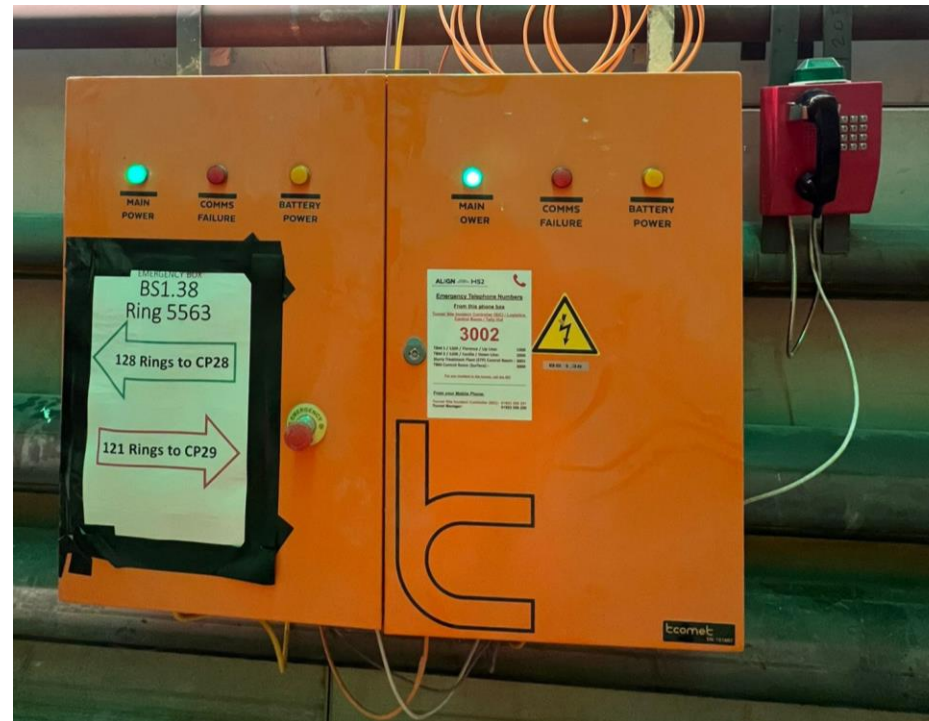
Emergency Boxes

There are emergency boxes every 300m of the tunnel. In an emergency they are used to raise the alarm and to contact the Site Incident Controller. They function even when the power and Wi-Fi is down.

How to Raise the Alarm

In an event of a fire or a major incident **PUSH THE BUTTON.**

This will sound the alarm and alert the Site Incident Controller.



Fixed Phone

In an emergency the Site Incident Controller can be reached by dialling **3002** (number is displayed on the emergency box).

The Phone can be used during a power loss and without Wi-Fi.

Lighting 'colour code' in tunnel

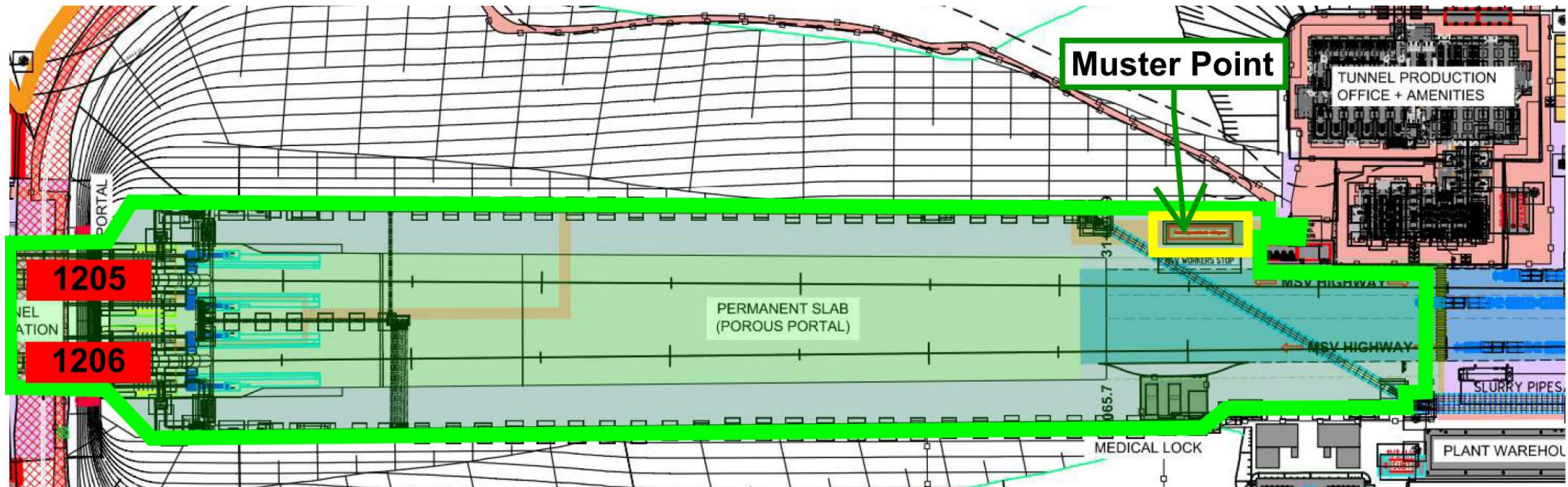


BLUE – Cross Passage complete, connected to another tunnel – Emergency exit route



GREEN – position of telephone and fire/emergency alarm activation button.

Muster Points – External Area

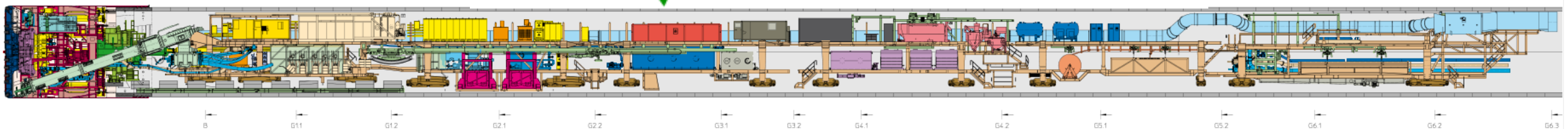


Muster Points – TBM

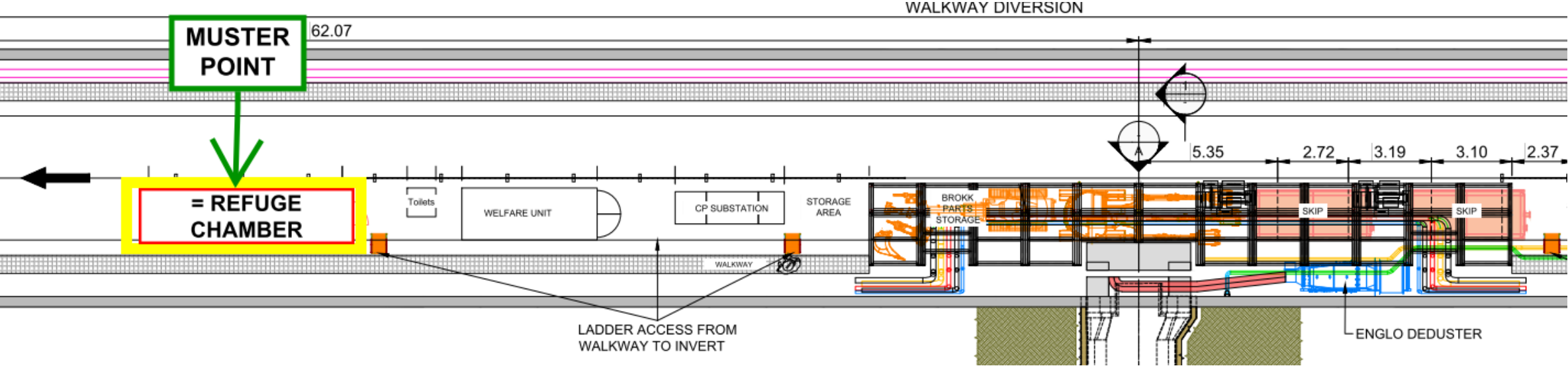


Muster Point

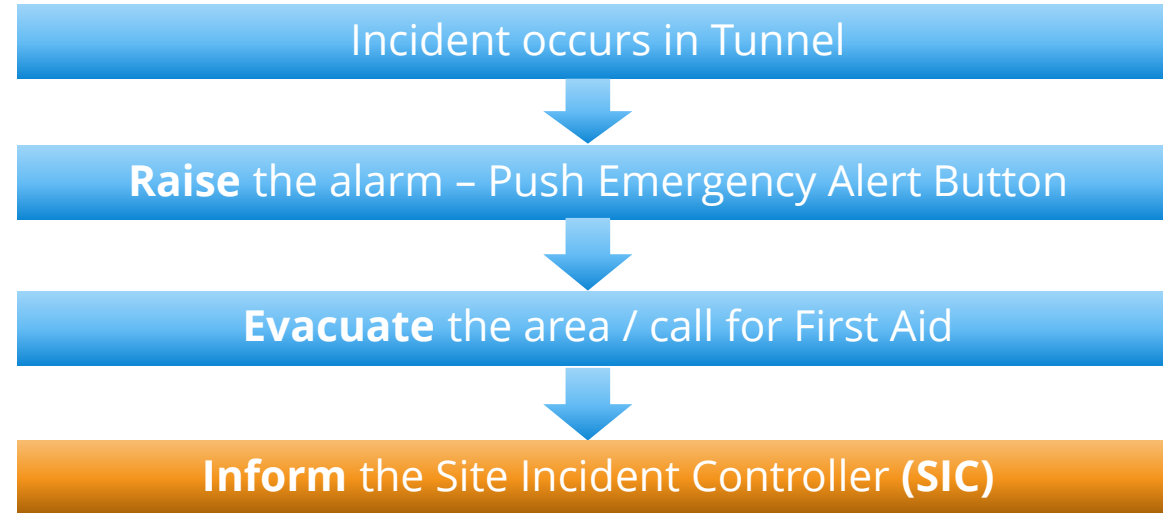
= REFUGE CHAMBER



Muster Points – CP



In Case of Emergency



Tunnel SIC number

From your phone:
01923 559 251 / 07826 510 289

From fixed phone:
3002

Every 300m

Self-Rescuer Donning Procedure



Questions?

Welcome to the Chiltern Tunnels

